The current issue and full text archive of this journal is available on Emerald Insight at:
https://www.emerald.com/insight/0957-4093.htm



Received 2 June 2024
Revised 12 December 2024
26 February 2025
Accepted 13 April 2025

# Evolving capabilities of nonprofit organizations in circular used clothing supply chains

Kanchana Dissanayake
*Department of Business Administration and Textile Management, Swedish School of Textiles, University of Borås, Borås, Sweden, and*
Rudrajeet Pal
*Department of Business Administration and Textile Management, Swedish School of Textiles, University of Borås, Borås, Sweden and Department of Industrial Engineering and Management, University of Gävle, Gävle, Sweden*

## Abstract

**Purpose** – Circular supply chains (CSCs) have experienced significant transformations, reflecting shifts in consumer preferences, a growing emphasis on the circular economy and increasing regulatory pressures. In the used clothing sector, nonprofit organizations (NPOs) have traditionally managed CSCs, but growing competition from for-profit entities is driving NPOs to adopt profit-oriented models. This paper examines how NPOs transformed their CSC capabilities to navigate this shift.

**Design/methodology/approach** – This study employs a case study approach to collect data from three NPOs, among Sweden's largest operators of used clothing CSCs, each with distinct operational structures. The dynamic capability view guides the design of semi-structured interviews and data analysis.

**Findings** – This study identifies seven microfoundations of dynamic capabilities crucial for the competitive management of used clothing CSCs. These seven microfoundations emerged across four distinct levels representing employee, managerial, organizational and supply chain levels. These findings provide a holistic framework for understanding and enabling competitive capabilities in functioning used clothing CSCs.

**Originality/value** – This study offers empirical insights into maintaining efficiency and competitiveness in used clothing CSCs. Although the evidence is drawn from NPOs, the findings also apply to for-profits, as NPOs have shifted toward profit-driven models. Additionally, this study guides for-profit and policymakers by promoting collaboration with NPOs to achieve synergies. Finally, this study advances the dynamic capability view by introducing four levels at which capabilities arise and their importance for competitive CSC management.

**Keywords** Circular supply chain, Used clothing, Nonprofit organizations, Dynamic capability

**Paper type** Research paper

Abbreviations

| | |
|---|---|
| CSC | Circular supply chain |
| DCV | Dynamic Capability View |
| EPR | Extended Producer Responsibility |
| NPO | Non-profit organizations |

## 1. Introduction

The transition of consumer goods supply chains towards circular supply chains (CSCs) is increasingly driven by regulatory measures such as waste disposal regulations, landfill bans,

The International Journal of Logistics Management
Vol. 36 No. 7, 2025
pp. 222-241
Emerald Publishing Limited
e-ISSN: 1758-6550
p-ISSN: 0957-4093
DOI 10.1108/IJLM-06-2024-0343

© Kanchana Dissanayake and Rudrajeet Pal. Published by Emerald Publishing Limited. This article is published under the Creative Commons Attribution (CC BY 4.0) licence. Anyone may reproduce, distribute, translate and create derivative works of this article (for both commercial and non-commercial purposes), subject to full attribution to the original publication and authors. The full terms of this licence may be seen at http://creativecommons.org/licences/by/4.0/legalcode

The authors acknowledge Sara Harper for her valuable assistance with language editing.
*Funding:* This research was supported by Familjen Kamprads Stiftelse (Grant No. 20200105).

Extended Producer Responsibility (EPR) schemes (Gupt and Sahay, 2015; Nithya *et al.*, 2021), and stakeholder pressures (Jakhar *et al.*, 2019; Lahane *et al.*, 2020). As these factors converge, managing CSCs requires businesses to develop dynamic capabilities that balance profitability with sustainability goals. Key enablers include building reverse logistics and circular supply chain expertise tailored to circular markets (Mishra *et al.*, 2023; Vegter *et al.*, 2020), fostering flexibility and innovation, and forging collaborative partnerships to generate new knowledge and enable resilience (Chari *et al.*, 2022; Coppola *et al.*, 2023). A successful CSC depends on an organization's ability to effectively structure these functions, often requiring new resources and cultivating dynamic capabilities (DCs) to adapt to changing demands and sustain competitive advantage (Meier *et al.*, 2023).

Nonprofit organizations (NPOs) play a dominant role in several reuse-based CSCs, forming crucial mainstays of the circular economy, particularly for textiles (Farahani *et al.*, 2022; Zhuravleva, 2024). NPOs have been the largest stakeholder of used clothing CSCs due to their unique set of process capabilities, such as organized collection channels, manual sorting, and trading. However, the need for a rapid transition to a circular economy, pressure from external stakeholders, and impending legislation such as EPR are changing the used clothing CSC landscape. Consumption of used clothes is growing in the "global north" as consumers are exploring more peer-to-peer and e-commerce sales (Persson and Hinton, 2023) while many for-profit actors such as fashion brands and retailers are making rapid entry into used clothing CSCs, creating a highly competitive environment (Adro and Fernandes, 2022; Zanjirani Farahani *et al.*, 2022). As a result, traditional actors such as NPOs face challenges in securing their position in used clothing CSCs, compelling them to develop DCs in managing competitive supply chains (Adro and Fernandes, 2022; Peng and Liang, 2019; Zhuravleva, 2024). However, existing research on DCs in CSCs is mostly dedicated to the for-profit sector (Chari *et al.*, 2022; Khan *et al.*, 2021; Ritola *et al.*, 2021) and the findings are less transferable to NPOs due to different organizational motivations and logic applied to NPOs such as inherent limitations on resources, strong commitment to social responsibility, and reliance on a volunteer workforce (Shevchenko *et al.*, 2024; Zhuravleva, 2024). To explore this interesting knowledge gap, this paper addresses the following research question:

> How do NPOs develop and deploy their capabilities to navigate the evolving landscape in used clothing circular supply chains?

This study is highly relevant given the increasing emphasis on circularity within the clothing industry, aiming to reduce resource consumption and minimize waste generation (Lahane *et al.*, 2020). NPOs often play a pivotal role in operationalizing the textile circular economy by collecting, sorting, redistributing, and reselling used clothes. NPOs bridge gaps where for-profit entities may lack incentives, focusing on value recovery that aligns with environmental and social goals. Their adaptive capabilities are critical for addressing logistical, social, and economic complexities in managing CSCs, and understanding how NPOS develop and deploy capabilities within CSCs can provide valuable insights for enhancing textile circularity.

The remainder of the paper is structured as follows. The literature review provides insights into the theoretical framework and explores how DCs are utilized in managing CSCs to enhance competitive edge. Next, we discuss the research methodology and present the results and discussion. Finally, we outline key conclusions and research directions.

## 2. Literature review

### *2.1 DC view*

The DC view emphasizes a firm's capacity to adapt to rapidly changing environments (Teece, 2007). This entails continually sensing, seizing, and reconfiguring resources and capabilities to sustain competitive advantage (Chatterjee *et al.*, 2022; Weaven *et al.*, 2021). DCs not only drive sustainable growth and strategic innovation but also reflect a balance between firm-specific path dependencies and common best practices (Moroni *et al.*, 2022; Qiu *et al.*, 2022).

Firms require DCs to anticipate, respond to, and recover from disruptions, enhancing performance, competitiveness, and supply chain resilience (Navarro-García *et al.*, 2024; Stadtfeld and Gruchmann, 2024). DCV can extend beyond for-profit organizations to NPOs (Chatterjee *et al.*, 2022; Weaven *et al.*, 2021), providing a solid theoretical foundation to explore the capabilities of NPOs in CSCs, however, there is a dearth of research in this field.

DCs can emerge across various organizational and hierarchical levels. Managerial DCs are vital for strategic decision-making (Heubeck, 2024) and helping firms utilize their employees, operations, and strategies to gain a competitive advantage (Bhadra *et al.*, 2024). Employee DCs are critical for identifying problems, generating new ideas, and implementing changes (Panagiotopoulos *et al.*, 2023). Organizational DCs are the capabilities embedded in the organizational structure, processes, technology, partnerships, and skills (Babaei and Aghdassi, 2022). Organizations should nurture these capabilities by developing managerial and employee capabilities through appropriate training and skill development activities (Widianto *et al.*, 2021). Furthermore, in navigating the volatile business environment, the cultivation of supply chain DCs becomes imperative to achieve competitive advantage and resilience (Chowdhury and Quaddus, 2017; Stadtfeld and Gruchmann, 2024). In recent literature, DCs have been widely discussed as a necessary factor for integrating triple-bottom-line sustainability into supply chains (Lu *et al.*, 2024). Regardless of their importance, there is a limited discourse on the significance of nurturing DCs across different levels and their influence on overall organizational performance.

### *2.2 DCs in the CSC context*

Emerging environmental, legislative, and stakeholder pressures are increasing the complexity and vulnerability of CSCs, necessitating the development of CSC-specific resources and capabilities to manage these complexities (Chari *et al.*, 2022). Due to the inherent uncertainties in CSCs, DCV provides a vital theoretical lens for organizations to extend, alter, and reconfigure existing operational capabilities into new capabilities that can help organizations better adapt to evolving environments (Bag and Rahman, 2023; Schöggl *et al.*, 2024). Microfoundations of DCs are unique to each organization, which enables organizations to achieve a competitive advantage by evolving and improving faster than competitors (Chari *et al.*, 2022). Additionally, DCs can facilitate a collaborative approach among stakeholders, further enhancing organizations' ability to adapt and thrive in a rapidly changing environment (Coppola *et al.*, 2023). Integration of the DC framework within CSCs not only addresses the challenges posed by environmental, legislative, and stakeholder pressures but also positions organizations to achieve sustainable competitive advantage through continuous innovation and adaptation.

Microfoundations of sensing combine a variety of actions and information sources, drawing from both internal and external exercises (Chari *et al.*, 2022; Khan *et al.*, 2020). Sensing supply chain disruptions can strengthen security and resilience by enabling proactive countermeasures (Stadtfeld and Gruchmann, 2024). Internal sensing involves supply chain managers working closely with team members to guide the sensing process and continuously monitor internal systems (Bag and Rahman, 2023). This internal focus is crucial for maintaining operational efficiency and addressing potential challenges. However, reliance on internal sources is not sufficient if organizations fail to recognize broader market shifts. This necessitates a balanced approach where internal monitoring is complemented by external insights. External monitoring involves market monitoring and technology scanning to identify market trends, competitor actions, new knowledge for developing novel products and processes (Khan *et al.*, 2020), and information sharing to explore potential collaborations (Meier *et al.*, 2023). However, the evolving nature of CSCs means that information can quickly become outdated, requiring a continuous monitoring approach. Collaborating with external stakeholders such as suppliers, customers, research institutes, and other industry actors further aids in external market updates, generating new ideas, and learning about innovative solutions (Coppola *et al.*, 2023; Khan *et al.*, 2020). Additionally, it is crucial to work closely with

customers to understand their priorities and to identify operational capabilities needed to meet customer demands and interests (Bag and Rahman, 2023). This requires a robust mechanism for gathering and analyzing customer feedback and learning from failures (Ritola *et al.*, 2021).

Microfoundations of seizing capabilities in CSCs emphasize the importance of constructing new logistics and supply chain structures, particularly through partnerships with local entities, as highlighted by Chari *et al.* (2022). This approach is crucial for fostering resilience and adaptability within CSCs. Additionally, Khan *et al.* (2020) argue that redesigning business models is fundamental to these capabilities, suggesting that innovation in business processes is pivotal for CSC's success. Integrating digital and advanced logistical capabilities significantly enhances CSCs by improving agility, flexibility, and responsiveness (Bag *et al.*, 2020). Moreover, establishing new partnerships within the supply chain is crucial for several reasons. Lu *et al.* (2022) and Bag and Rahman (2023) highlight the importance of these collaborations for accessing raw materials and customers, better management of reverse logistics, sharing responsibilities for product recovery, and realizing new business models. Enhancing communication and transparency along the supply chain is another aspect that can significantly improve the overall performance of CSCs (Quayson *et al.*, 2023). Finally, effective change management strategies are essential to support seizing opportunities to mitigate employee resistance when implementing new systems and processes (van Eechoud and Ganzaroli, 2023).

Reconfiguring involves effectively recombining and realigning resources to gain a competitive advantage in CSCs for pursuing long-term success (Bag and Rahman, 2023; Sandberg and Hultberg, 2021). This process requires reconfiguring technological, environmental, and human capabilities to build circular and resilient supply chains within the evolving landscape (Chari *et al.*, 2022). Literature reveals varying perspectives on the implementation of these strategies. For instance, Khan *et al.* (2020) identify restructuring, technological upgradation, knowledge integration, and the adaptation of best practices as essential microfoundations of reconfiguring capabilities. Bag and Rahman (2023) suggested that achieving flexibility in CSCs in terms of product, volume, sourcing, delivery, and markets is essential for survival in the uncertain and competitive landscape. Moreover, continuous updates in knowledge regarding markets, competitors, and technologies are essential to reconfigure CSCs accordingly (Quayson *et al.*, 2023; Sandberg and Hultberg, 2021). By focusing on these capabilities, organizations can ensure their CSCs are not only resilient but also capable of adapting to new challenges and opportunities, thus securing a long-term competitive advantage.

## 3. Methodology

A qualitative case study approach was chosen to explore microfoundations of DCs in NPOs engaged in used clothing CSCs. Case studies offer detailed insights into specific contexts (Yin, 2013), which are essential for understanding the intricacies of DCs within NPOs, and facilitating both theory elaboration and expansion of theories (Fisher and Aguinis, 2017). Abductive reasoning, employed in this study, proves valuable for exploring this relatively unexplored area. Abduction enables flexible interpretation of observed facts, allowing for the generation of the best possible explanation beyond what is already known (Kovács and Spens, 2005).

### *3.1 Case selection*

This study utilized a multiple-case study approach chosen for its ability to examine a phenomenon across diverse contexts, facilitating comparisons, and enhancing the generalizability of findings (Gustafsson, 2017; Yin, 2009). Case selection involved a combination of theoretical and purposive sampling, guided by the principle of selecting information-rich cases that could provide valuable insights (Patton, 2015). In Sweden, NPOs dominate the used clothing CSCs, managing all key operations such as collections, sorting,

distribution, and resale. This makes them particularly suitable for this study. These NPOs operate under three distinct operational structures: centralized, semi-centralized, and decentralized, which formed the basis for theoretical sampling. Case samples were selected to represent each structural type as outlined in Table 1. Additionally, purposive sampling was applied to identify three representative case organizations among the largest NPOs in Sweden involved in used clothing CSCs, chosen for their scale and scope. Their comprehensive engagement across all stages of the supply chain and willingness to share operational data further justified the selection. The decision to include only three organizations was guided by the need to balance the depth and breadth of analysis while ensuring the study remained manageable. This selection ensures that the study captures the diversity and complexity of used clothing CSCs while addressing the critical capabilities and challenges central to this phenomenon.

### *3.2 Data collection*

Data collection encompassed three sources: interviews, site visits, and document analysis, as outlined in Table 2. Primary data were gathered through open-ended interviews with case organizations, employing a semi-structured questionnaire developed based on the initial literature review. Mainly, "how" questions were included to invite participants to provide deeper insights into their experiences. The questionnaire and the purpose were communicated in advance to obtain consent from the organizations. A total of 25 interviews were conducted, 9 with NPO1 and 7 each with NPO2 and NPO3. Initially, top management was interviewed, followed by a snowball approach to identify key individuals responsible for the supply chain operations. Interviews, each lasting about 30–60 min, were recorded with permission and later transcribed. Additionally, site visits facilitated a comprehensive understanding of the process

**Table 1.** Case organizations and activities

| Activities | NPO1 | NPO2 | NPO3 |
|---|---|---|---|
| Collection methods | In-store donations<br>Home collections | Collection bins (2,500 bins)<br>Multiple waste sites (200 containers)<br>Retailers' take-back programs<br>Other NPOs | In-store donations.<br>Retailers' take-back programs<br>Home collections |
| Collection volumes | 432 tons (data available only for 2 hubs) | 13,000 tons | 1,500 tons |
| Sorting facilities | Semi-centralized<br>Sorting is carried out in 4 centralized sorting hubs and in individual stores that are not connected to sorting hubs | Decentralized<br>Sorting is carried out in 3 regional sorting centers | Centralized<br>Sorting is carried out in the central facility |
| Sorting strategy | 100% of the collected volume is sorted | Only around 6% of the collected volume is sorted | 100% of the collected volume is sorted |
| Distribution strategy and volumes | Export −35%<br>Sold in Sweden −15%<br>Energy-32%<br>Recycling-16%<br>E-commerce-2% | Export ~80%<br>Sold in Sweden <1%<br>Recycle- 6%<br>Rags/wipes – 8%<br>Incineration – 4%<br>Non-textiles- 2% | Export-34.5%<br>Sold in Sweden – 64%<br>Remake and waste – 1.5% |
| Number of local shops | 61 physical stores | 11 physical stores | 25 physical stores and 2 web shops |

**Source(s):** Authors' own work

**Table 2.** Data sources

| Organization | Interviews | Site visits | Documents |
|---|---|---|---|
| NPO1 | 9 interviews<br>- Sustainability manager<br>- Logistics manager<br>- Store managers | 3 visits to a centralized hub<br>1 visit to a concept store- | Sustainability reports<br>Company website<br>Online materials |
| NPO2 | 7 interviews<br>- Business director<br>- Deputy director<br>- Logistics manager<br>- Regional manager | 1 visit to the headquarters and the warehouse | Annual reports<br>Company website<br>Online materials and videos |
| NPO3 | 7 interviews<br>- Area manager production<br>- Area Manager Sales<br>- Team manager<br>- Administrator<br>- Designer | 1 visit to the main facility<br>1 visit to the remake facility<br>1 visit to a store | Company website<br>Online materials<br>Published articles |

**Source(s):** Authors' own work

flow, resource allocations, limitations, and organizational capabilities. Notes were taken for reference and cross-verification of interview data. Supplementary documents addressing organizational processes and activities were collected to enhance the understanding of the organizational processes and capabilities, and the readiness to confront external challenges.

### *3.3 Data analysis*

The data analysis followed a qualitative manual content analysis as outlined by Renz *et al.* (2018). This process involved reading transcripts repeatedly, making notes to categorize different types of information, developing a coding scheme to organize data, and identifying emerging themes. As elaborated in Figure 1, an abductive reasoning approach guided the data coding, beginning with *a priori* categorization based on DC theory, followed by open and axial coding to label and merge repetitive codes while developing new themes (Blair, 2015). The interview data were categorized deductively based on the three core elements of Dynamic Capabilities theory, which also shaped the interview questions: sensing (identifying challenges and opportunities), seizing (taking action), and reconfiguring (adapting and realigning resources).

In the next step, an inductive approach was used to develop codes for the data within each category. The interviewees' responses to "how" questions guided this open coding process, specifically how organizations engage in sensing, seizing, and reconfiguring. This involved moving back and forth between data and theory and creating a coding scheme that organizes data into a more structured framework that represents patterns. Narratives further justified the code assignment, and constant comparison refined the code category. This step led to identifying recurring patterns and first-order categorization of results, as shown in Figure 1. Moving to the second-order category, axial coding identified relationships among categories, creating a comprehensive second-order categorization representing emerging themes. These themes were defined to provide the best possible explanations of microfoundations discovered in this study. This approach was particularly helpful in refining the understanding of the microfoundations of sensing, seizing, and reconfiguring activities.

Sorting and coding data followed a structured procedure ensuring inter-coder reliability (Seuring and Gold, 2012). Each interview was initially categorized and coded by the corresponding author by continuously verifying against established codes while incorporating

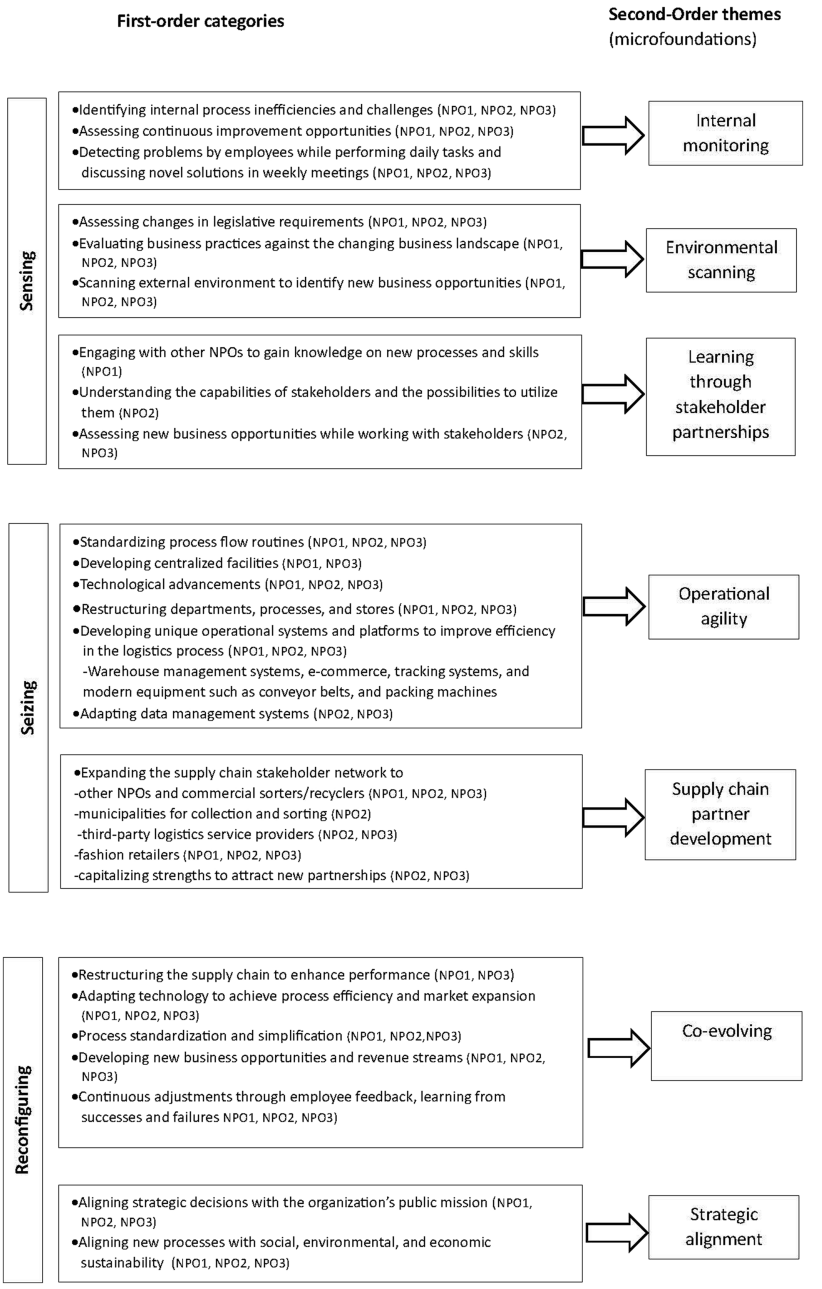



**Figure 1.** Data structure. Source: Authors' own work

new data, to avoid shifts in meanings over time. Cross-validation of the coded data by the co-author ensured consistency and reliability. The coding process was repeated until both authors agreed and confirmed new patterns and themes, further enhancing the inter-coder

reliability. This iterative process resulted in a structured set of coded data that could be categorized, compared, and subjected to analysis.

*3.4 Research quality*

The research methodology in this study was meticulously designed to ensure reliability and validity. Data triangulation was ensured by implementing multiple data collection procedures, as described by Renz *et al.* (2018). This rigorous data collection procedure included interviews, site visits, and published documents to enhance the reliability and depth of findings. To ensure consistency across data sources, Interview data were cross-checked against field notes and secondary documents, allowing for the validation of key insights and minimizing potential biases (Thyer, 2010). For instance, statements from interviewees regarding technology adoption were compared with direct observations made during site visits. This comparative approach ensured that the interpretations remained aligned and reinforced the credibility of the study's conclusions. This process of data triangulation enhances internal validity (Yin, 2009) by confirming findings across multiple sources and minimizing the risk of bias or misinterpretation. Additionally, each step of the data collection process built upon the previous one, utilizing snowball sampling to obtain comprehensive data and ensure the consistency of previously gathered data. The inclusion of diverse data sources and multiple case studies enhanced generalizability and external validity. Overall, the methodology employed data triangulation techniques, systematic coding, and iterative data collection to ensure reliability, while promoting transparency and reproducibility.

## 4. Results

The evolution of microfoundations of sensing, seizing, and reconfiguring represents fundamental building blocks that allow NPOs to effectively manage their supply chains within a dynamic and complex environment. Unique challenges and opportunities faced by NPOs in this specific environment contribute to the evolution and refinement of these microfoundations, providing a nuanced understanding of how they can effectively sense, seize, and reconfigure for sustained success in a dynamic landscape. This section underscores the importance of seven microfoundations, based on the findings from the case studies conducted with three NPOs.

*4.1 Microfoundations to sensing DCs*

NPOs' ability to sense both the internal and external environment was identified through three microfoundations: *internal monitoring*, *environmental scanning*, and *stakeholder engagement*.

*Internal monitoring* emerged as a fundamental microfoundation of sensing DCs. Our NPOs actively engage in internal and continuous learning, where internal operations are frequently assessed to identify challenges and potential improvement opportunities. All NPOs demonstrate a proactive approach to identifying and mitigating challenges within used clothes CSCs. NPO1 and NPO3 identified decentralized operations as a major inefficiency, causing bottlenecks in logistics, such as sorting and distribution. NPO2's monitoring process also recognized ongoing challenges in logistics operations, such as inefficiencies in collection and transportation.

> We have to know the right information at the right time, for example, this container will be full on Wednesday we need to have an empty one on Wednesday, and the collection is not going as planned, so it's only half full, and the transport is already booked, and the truck is on the way. (Logistics Manager-NPO2)

Internal monitoring relies heavily on feedback from employees and management as they engage in daily activities. This ongoing exchange of information facilitates a dynamic grasp of

operational challenges, enabling timely improvements. Nonetheless, frequent changes in the workforce within NPOs pose challenges in maintaining process efficiency, yet provide an opportunity to receive feedback and innovative ideas from fresh minds, in contrast to a traditional organization with a stable permanent workforce.

*Environmental scanning* was noticeable among all NPOs as they possess a keen sense of environmental dynamics affecting used clothing CSCs. This includes monitoring changes in their external market demands, regulatory environments, technological advancements, and social trends related to used clothing CSCs. All NPOs are attuned to external challenges such as upcoming legislative changes, including EPR schemes, an increasing influx of low-quality clothes, and growing competition from for-profit entities such as commercial recyclers and fashion brands. In response to this increasing competition and legislative changes, they prioritize exploring and adopting technological advancements such as e-commerce, efficient logistics, inventory management systems, and modern equipment to optimize sorting and distribution. For instance, NPO2 currently exports over 90% of textiles without sorting, but with EU legislation impending, restricting the export of "textile wastes", NPO2 aims to incorporate pre-sorting into operations.

> We have to turn waste into products. What we are convinced is that they (the EU) will impose rules that you cannot export waste. It will have to be sorted before you send it out. This is our way (planning pre-sorting) to prepare for that legislation. (Deputy Director-NPO2)

*Learning through stakeholder engagement* was utilized by NPOs to learn from the best practices of other stakeholders. NPO1 took a proactive approach by engaging with Oxfam, a prominent UK-based NPO, focusing on delving into the intricacies of e-commerce activities and understanding the process requirements.

> We had a nice corporation then and still have with Oxfam in England. Oxfam was already doing something similar. So, I know that a group from (NPO1) went to England to make a visit and study how are they doing this. (Logistics Manager-NPO1)

Similarly, NPO3 sought insights from the largest online platform in the Nordic region dedicated to selling used clothes. By leveraging the expertise of this successful online platform, NPO3 aimed to develop its e-commerce platform, aligning its strategies with the dynamic landscape of online commerce. Meanwhile, by engaging with overseas sorting facilities, NPO2 sensed an opportunity to address the skill deficit in sorting used clothes for reuse in Sweden.

### *4.2 Microfoundations of seizing DCs*

This study identified two microfoundations of seizing capabilities as *operational agility* and *supply chain partner development*, as elucidated below.

*Operational agility*, as demonstrated by NPO1, involved developing centralized sorting hubs to enhance resource and logistics efficiency. Centralization allowed effective monitoring of sales and stock levels of stores connected to the hub, facilitating efficient stock replenishment. Process efficiency of the centralized hub was further enhanced by developing a distinctive warehouse management system, streamlining the storage, retrieval, and distribution of unique, one-off items, addressing a significant logistical challenge associated with used clothes. Moreover, NPO1 streamlined procedures for receiving, sorting, and pricing used clothes, aiming to enhance process efficiency and facilitate quick learning for new employees. Facilitating quick learning is particularly crucial for NPOs experiencing a frequent rotation of the workforce as part of work integration activities. Additionally, NPO2 incorporated technology and modern equipment to enhance logistics efficiency in the collection, sorting, and packing processes. QR codes and mobile apps were introduced to track fill rates of collection bins, enabling real-time reporting and efficient organization of collection trucks, and optimizing transportation routes.

> When the container is full, we report it in our inbuilt application in our phones, so they just scan a QR code, we also know how much is inside in the collection boxes in the cities or around the municipality, we also have a QR code in that bin. (Logistics Manager-NPO2).

In response to upcoming legislative changes that would prevent the export of textile "waste", NPO2 implemented a pre-sorting process by incorporating an automated conveyor belt. This process aimed at efficiently removing non-textile items and waste, thereby preparing them as usable products in export markets. Additionally, NPO1 and NPO3 developed e-commerce platforms to become competitive in sales platforms and to enhance profitability.

*Supply chain (SC) partner development* is a crucial microfoundation demonstrated by NPOs, especially in response to EPR and heightened market competition. NPO1 seeks new partnerships and collaborations with research institutes and commercial sorters/recyclers, aiming to reestablish its position within the changing environment. However, negotiations with recyclers for long-term partnerships disclosed challenges, where recyclers demand high-quality sorted feedstock based on material compositions, and NPO1 does not have the skills in the labor force to carry out sorting for recycling at that granular level. In response to this challenge, NPO1 is currently investigating its capabilities to become a sorting hub for recycling. NPO2, on the other hand, strategically expands its collection through partnerships with real estate agencies, municipalities, and logistics service providers. Having identified the challenges of carrying out high-quality sorting operations in Sweden, NPO2 collaborates with overseas sorting partners to get clothes sorted for resale in Sweden. Additionally, NPO2 positioned itself as a strategic logistics partner and invited potential collaborators to jointly tackle impending legislative challenges. NPO3 is leveraging the legislative challenges to engage with fashion retailers to obtain consumer returns, partnering with multi-brand retailers to sell their "Remake" collection, and promoting remaking expertise for new collaborations.

> Many companies approach us are saying we want to cooperate with the Remake . . . we have 3 or 4 small shops that buy our brand and mix with other brands, they buy clothes from us at a wholesale price . . ..we have an interest from bigger, like chain stores . . . (Designer-NPO3)

### *4.3 Microfoundations of reconfiguring DCs*

Key microfoundations of reconfiguring capabilities of the NPOs, as observed through our study, are *co-evolving* and *strategic alignment*.

*Co-evolving* capabilities of NPOs exemplify a change to the traditional SC framework and gear more toward a profit-driven used clothing CSC. For instance, NPO1 has undergone a significant restructuring of its operations by establishing centralized sorting hubs, developing a new warehouse management system, streamlining process routines, and revamping traditional stores using visual merchandising techniques. NPO2 invested in technology and modern equipment to remain responsive to changing market dynamics. Moreover, it has diversified new CSCs beyond used clothes, creating a new revenue stream by selling organic and recycled textile products to capitalize on growing market demand. Similarly, NPO3 creates a novel business model that remakes used clothes into new fashion pieces sold through an exclusive webshop. These adaptations are accompanied by feedback loops aimed at monitoring the performance of new systems and processes, consistently questioning their suitability, capturing lessons from experience, and continuously restructuring the processes and supply chain accordingly. For instance, despite initially venturing into e-commerce to broaden its consumer base, NPO1 encountered logistics challenges and high costs. Through ongoing monitoring and feedback, NPO1 opted to discontinue its e-commerce and instead collaborate with a third-party e-commerce platform.

> . . . if you, as a customer, buy more than one item in one order, and one item is in Lund and one item is here (Borås), there will be two packages. [. . .] We have to see how that affects our costs for selling products if the customer will not be interested in paying for two shipments, [. . .] So how do we cover that cost, and how do we make it still work? (Logistics Manager-NPO1).

*Strategic alignment* is recognized as a crucial microfoundation for all NPOs in navigating complex landscapes without impeding social impact. As NPOs transition towards profit-oriented business models, they demonstrate the capability of restructuring operations while accommodating social needs, which is paramount. NPO2's strategic collaborations with social enterprises and municipalities for logistics services and sorting operations exemplify its reconfiguring capability, aligning with the governmental mission to address unemployment and foster community support. Similarly, NPO1 places a strong emphasis on achieving equilibrium among the "three bubbles"- social, economic, and environmental sustainability when restructuring operations and processes within used clothing CSCs. There is a strong focus on evaluating the impact on employees and overall social sustainability when implementing new systems and functions, aiming to achieve a harmonious balance between economic and social sustainability.

> . . . We build functions and functions and functions, and no one really stops to say, but why? Why do we really need this? [. . .].So we had to always think, yeah, nice to have and need to have stuff, do we really need this function? How will it make it easier for the employees to work with the system? If we can't answer how it's better for the employees, but better for the profit, we probably shouldn't have it. (Logistics Manager-NPO1)

*4.4 Enablers of microfoundations across different levels of origin*

The seven microfoundations of DCs identified in this study play a crucial role in helping NPOs to competitively reposition their used clothing CSCs within the changing landscape. Understanding how these DCs emerge requires an in-depth examination of the underlying factors that drive their development and deployment. This section is devoted to a second-level analysis to explore how the seven microfoundations identified in this study manifest across different levels of organization. Moreover, breaking down the distinct contributions and interactions at each level facilitates identifying key enablers that promote the growth of DCs at each level, offering insights into how NPOs can better align their strategies, structures, and processes to foster the adaptability and resilience of used clothing CSCs. The following analysis seeks to provide a comprehensive view of the levels of emergence of DCs and mechanisms that support the evolution of DCs within their used clothing CSCs.

The ability to sense, particularly through *internal monitoring*, predominantly emerged as a *managerial-level DC* derived from their knowledge and operational skills. However, this activity is complemented by *employee-level DCs* to grasp the process-level challenges and identify opportunities for improvement. Notably, in the context of all NPOs, encouraging feedback from employees regarding the process efficacy and brainstorming new improvement ideas has been instrumental in enhancing *internal monitoring*. This is further evident within NPO1, as it benefits from the frequent rotation of the workforce as a part of work integration activities, facilitating a constant influx of fresh perspectives and ideas.

> . . . there are also a lot of good questions (from employees) and a lot of why, why don't you do it like this? Why is it done like that? And that makes us also have to think all the time.(Logistics Manager-NPO1)

In contrast, in the case of NPO2, reliance on municipality-managed work integration activities for textile sorting operations has posed challenges due to the pursuit of two divergent goals. Utilizing unemployed workers without having sorting skills, interests, or fashion sense results in low expectations for *employee-level DCs* in sensing. Consequently, *managerial-level* DCs become crucial in the sensing stage. Other microfoundations of sensing – *environmental scanning* and *learning through stakeholder partnerships*– represent *organizational-level* DCs, although their effectiveness is contingent upon managerial involvement in the learning process and their abilities to grasp opportunities. Cases show evidence that NPOs at the *organizational level* pay close attention to the external environment and evolving landscape of used clothing CSCs, including impending legislation and growing competition from commercial entities.

NPO1 and NPO3 actively engage with external stakeholder organizations to learn from their processes and systems.

*Operational agility* in seizing, primarily an *organizational-level DC*, is deployed within the case organizations through technology adoption, improving logistics efficiency, integrating information systems for data-driven decision-making, and incorporating best practices learned from stakeholders. Legislative changes also act as catalysts, propelling all NPOs to enhance *operational agility* to remain competitive within the changing landscape of used clothing CSCs. Notably, *operational agility* is significantly influenced by *managerial-level DCs*, particularly in the development of new processes and systems, as they oversee the change management process. However, NPO1 realized that the middle management often operates as followers and occasionally exhibits resistance to change.

> When we come to Lund, we say, oh, we tried this in Borås, it was really great, and, they could be like, yeah, but it will never work here (Logistics Manager-NPO1).

Moreover, *SC partner development* stems from the *SC-level* and is influenced by the *organizational-level DCs* because *organizational-level DCs* are pivotal in the selection of new actors as supply chain partners. However, NPO2 admits the challenges of developing new partnerships along the supply chain, specifically due to their presence as NPOs and having conflicts of interest and divergent goals when engaging with for-profit entities. NPO2, on the other hand, discovered issues in partnering with municipalities to sort textiles because of the lack of upskilled labor and mismatched interests. Fashion brands attempted to utilize the "remake" capabilities of NPO3, however, NPO3 realized that some fashion brands merely sought to employ this as a marketing strategy without genuinely achieving environmental benefits.

In the reconfiguration stage, *co-evolving* primarily resides at the *organizational-level DC*. However, *SC-level DCs* are crucial in both the development and deployment of these microfoundations, because reconceptualization of the supply chain plays a significant role in *co-evolving*. For instance, NPO2 expanded its supply chain by partnering with overseas sorting partners and sustainable textile producers to develop new business models and revenue streams.

> We actually buy clothes from our customers (overseas). They sort Swedish clothes, and we buy them back. [. . .] Actually we are trying another new business model which is quite interesting. [. . .], we are going to buy cotton sheets produced in India from recycled fibres and organically produced (Deputy Director-NPO2).

*Strategic alignment* becomes a crucial *organizational-level DC* for NPOs during the reconfiguring stage due to their dual mission of maintaining social sustainability while achieving economic sustainability within used clothing CSCs. This aspect is notably evident with all NPOs. For instance, NPO1 highlights the importance of balancing social, economic, and environmental aspects in its strategic decision-making process, ensuring that profitability never takes precedence over social sustainability. Similarly, NPO2 has implemented various strategic decisions to enhance its logistics management process, however, governmental unemployment programs were integrated into collection and sorting operations, even though employing unskilled workers may potentially negatively impact process efficiency.

Overall, Figure 2 elaborates on the interplay between the seven microfoundations across four levels and their enablers at each level, as identified through the study.

## 5. Discussion

The study findings demonstrate critical capabilities NPOs have developed and deployed in response to the evolving landscape of used clothing CSCs. Notably, NPOs have demonstrated their ability to effectively respond to impending legislation and growing competition from for-profit entities such as commercial recyclers and traditional fashion brands (Zhuravleva, 2024).

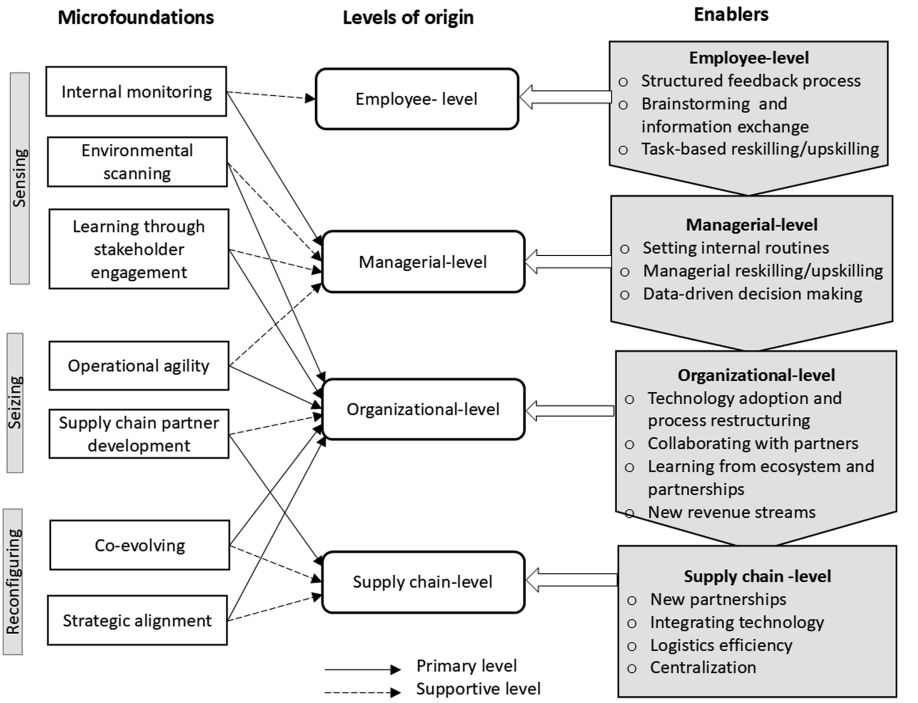



**Figure 2.** The framework of origins and enablers of microfoundations across levels. Source: Authors' own work

Within this context, *environmental scanning* emerges as a crucial microfoundation for NPOs to stay informed of external pressures, explore subsequent implications, and proactively restructure their supply chains to remain relevant and competitive. Additionally, the used clothing market is undergoing a rapid commercialization process, as stated by Persson and Hinton (2023), and therefore, *internal monitoring* and *learning through stakeholder engagement* have played key roles in identifying commercial strategies to align external demands with internal processes and structures.

This study emphasizes the importance of efficient logistics management facilitated by technological advancements as a vital capability to gain a competitive edge in used clothing CSCs. For instance, NPO2's utilization of technology-driven reverse logistics capabilities such as QR codes and mobile apps has proven instrumental in gathering information from diverse waste collection points. As emphasized by Stadtfeld and Gruchmann (2024), these types of scanning technologies can improve monitoring, visibility, and information reliability, while facilitating well-informed decision-making in the reverse logistics process (Bag *et al.*, 2020). Prompt and accurate information sharing can reduce uncertainties in logistics operations and facilitate *operational agility.* Furthermore, the complexities of managing used clothing CSCs stem from the unique nature of each garment, posing challenges in inventory management. NPO1 overcame this challenge by developing a unique warehouse management system tailored to handle individual pieces, thus optimizing logistics efficiency. This aligns with recent literature, highlighting efficient logistics management as a key enabler of *supply chain-level DCs*, requiring complex configurations of locations and capacities (Meier *et al.*, 2023).

Additionally, the study highlights the significance of *SC partner development* in addressing challenges within a competitive environment. Collaborative efforts enable organizations to capitalize on existing capabilities of partners, rather than duplicating efforts, and jointly

develop new skills and competencies (Polater, 2021). These new supply chain partnerships in used clothing SCs can enhance collection efficiency, foster collaborative sorting and reselling, drive knowledge sharing, and support scalable growth (Pal and Sandberg, 2024). Although NPOs demonstrate capabilities in developing these new SC partnerships, challenges persist, particularly due to conflicting interests arising from divergent goals among partners. NPO2 successfully deployed its capabilities to expand collections by partnering with municipalities and housing companies; however, collaborating with municipalities to utilize an unemployed or volunteer labor force for sorting introduced challenges regarding required skill levels and deliverables. As noted by Zhuravleva (2024), their learning and experiences are no more valuable when that temporary employment ends. NPO1 attempted to establish partnerships with commercial recycling companies, however, collaboration was hindered by recyclers' demands for sorted feedstock for recycling, exacerbated by the lack of an upskilled labor force to sort textiles for recycling.

This study has also revealed that NPOs are *co-evolving*, moving away from traditional operational frameworks and embracing elements of profit-oriented supply chains. This transformation involves enhancing reverse logistics capabilities through technology integration, improving process efficiencies, and diversifying product portfolios. This shift is largely influenced by the growing presence of for-profit entities within used clothing CSCs, prompting NPOs to prioritize profitability to maintain their position (Persson and Hinton, 2023). However, it is important to note that all NPOs emphasized the importance of *strategic alignment* in their actions to increase profits, ensuring that profit does not compromise the social sustainability of used clothing CSCs. Aligning with this finding, Shevchenko *et al.* (2024) observe that excessive focus on cost and operational efficiency can hinder NPOs from achieving their social goals.

Our study reveals that the seven microfoundations of DCs emerge across four distinct levels. *Managerial-level DCs* emerge as the most critical DC within the operational context of NPOs, as they are the driving force and change agents within competitive used clothing CSCs. We found the management exhibits a commendable capacity for making well-informed strategic decisions rooted in forward-thinking, resulting in the development of *organizational-level DCs*, as highlighted by Heubeck (2024). However, the *managerial-level DCs* are largely influenced by the manager's hierarchical level (Bish and Becker, 2016), as evidenced by NPO1, where middle management showed some resistance to change when seizing opportunities. Additionally, *Employee-level DCs* in problem detection and providing feedback for new processes were particularly valuable for managers in identifying challenges and opportunities for improvement. M*anagerial-level DCs* are significantly shaped by *employee-level DCs*, as evidenced in this study. Consistent with this finding, previous studies highlight the crucial role of employees in shaping managerial and organizational dynamic capabilities (Wohlgemuth *et al.*, 2019). However, considering that NPOs often engage volunteers and participants in work integration activities, and their misalignment of objectives, as evident in NPO2, revamping their training programs to reskill and upskill employees to incorporate the evolving dynamics in used clothing CSCs can benefit both employees and organizational success. Otherwise, NPOs, which often rely on volunteer workforces pursuing different needs and expectations, place an additional emphasis on capable managers (Bish and Becker, 2016).

*Supply-chain level DCs* are emerging, enhancing NPOs' responsiveness to market changes, which is demonstrated as operational agility by Rashid *et al.* (2024). However, challenges persist, notably arising from conflicting interests stemming from divergent goals among partners. This challenge became apparent during efforts to develop partnerships with both for-profit entities and municipalities, as extensively discussed in Section 4. However, to operate used clothing CSCs in a mutually beneficial manner, all stakeholders must assess opportunities to establish effective partnerships and synergies by utilizing each other's capabilities, rather than duplicating efforts. This necessitates engaging in constructive dialogue and respecting each other's capabilities and business goals.

*5.1 Theoretical implications*

This study makes theoretical contributions by extending the know-how of the application of DCs in enhancing competitiveness in the emerging field of CSC (Bag *et al.*, 2019; van Eechoud and Ganzaroli, 2023; Khan *et al.*, 2020). First, this study introduces an integrated framework that advances the theoretical understanding of DCs by categorizing their emergence across four distinct levels: employee, managerial, organizational, and supply chain, as shown in Figure 3. While previous studies examined DCs within some of these levels in isolation, our study provides a holistic perspective by integrating all levels, emphasizing the interdependence and alignment of DCs across levels. This integrated view shifts the theoretical discourse toward recognizing the systemic nature of DCs, emphasizing that their effective deployment requires coordinated efforts across multiple layers within and beyond the organization.

Secondly, this study significantly contributes to the limited discourse on enabling DCs by offering insights into how these capabilities can be nurtured across all levels. By identifying enablers at each level, the study bridges the gap between theoretical constructs and practical application. This framework serves as a foundational tool for further research, enabling scholars to explore the interplay of enablers and microfoundations across levels. Furthermore, the identification of enablers contributes to advancing theory by offering a structured approach to understanding how organizations can strategically develop DCs to enhance their resilience, adaptability, and competitiveness in CSCs.

Thirdly, this study enriches the literature on managing CSCs within uncertain and evolving contexts by providing an empirical lens to explore DCs in the specific domain of CSCs. While previous studies primarily focused on the CSC capabilities of for-profit actors, this study fills the gap by focusing on NPOs, leading to the identification of seven capabilities in the used clothing CSCs. In essence, this study redefines the theoretical boundaries of CSCs by offering new constructs, integrating multi-level insights, and providing a roadmap for future research to deepen the exploration of CSCs.

*5.2 Managerial implications*

This study offers valuable insights and practical guidance for organizations involved in CSCs to effectively sense opportunities, capitalize on them, and reconfigure their resources and

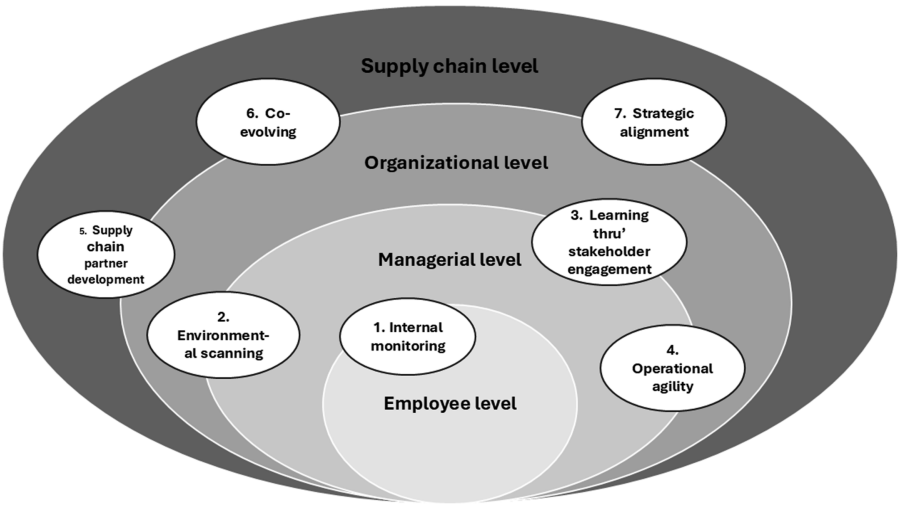



**Figure 3.** Deployment of capabilities across levels. Source: Authors' own work

processes to adapt to changing environmental conditions. Organizations can leverage the frameworks presented in Figures 2 and 3 to assess their existing capabilities, identify gaps, determine the requirements for developing new capabilities, and evaluate their ability to deploy them effectively across levels.

Figure 2 provides a practical framework for organizations to identify and evaluate the origins and enablers of microfoundations across managerial, employee, organizational, and supply chain levels. This framework is useful for organizations to identify and build the capabilities needed for effective CSC management. For example, at the employee level, task-based reskilling and structured feedback ensure staff are equipped with the skills to handle day-to-day issues. Managers can implement data-driven insights to refine operations, enabling faster adaptation to CSC demands. Effective logistics management, highlighted as a core capability in this study, relies on timely and accurate information flow to minimize uncertainties and facilitate robust decision-making. At the organizational level, adopting relevant technologies and collaborating with partners can streamline reverse logistics and reuse, creating cost-effective solutions. Supply chain-level actions, such as forming strategic partnerships and centralizing logistics, can improve coordination and scalability. By applying this framework, organizations can systematically evaluate their capabilities across four levels, identify gaps, and prioritize improvements aligned with their critical needs.

Figure 3 provides a useful framework for organizations to understand the deployment strategies of seven capabilities across four levels, showing the interconnections across these four levels. For example, internal monitoring links managerial and employee levels by translating high-level goals into actionable insights for day-to-day operations, fostering accountability and transparency. Similarly, environmental scanning at the managerial and organizational levels creates a shared understanding of external opportunities and risks, which drives strategic decisions and resource allocation. Stakeholder engagement and operational agility bridge the managerial and organizational levels by encouraging collaboration and adaptability. These overlapping areas reinforce cohesion across levels, ensuring that efforts at one level amplify those at another. By deploying these capabilities, organizations can improve their CSCs more effectively by better decision-making to manage challenges.

### *5.3 Societal implications*

The findings of this study highlight important social implications for sustainable development. NPOs provide affordable clothing options for low-income individuals while increasingly attracting consumers across economic backgrounds. This shift redefines local and regional consumption, promoting sustainable lifestyles. Additionally, used clothing CSCs have significant potential to generate employment, as they rely heavily on labor-intensive activities. Job opportunities are prevalent in key operations such as collection, sorting, distribution, and reselling. Interviews and site visits further revealed that NPOs employ a diverse range of individuals, including the elderly, pensioners, migrants, people with disabilities, and other vulnerable groups, contributing to job creation and supporting livelihoods. Addressing skill gaps identified in this study requires developing employee-level DCs through upskilling and reskilling. Inclusive training programs for migrants and economically marginalized groups can prepare them for both white- and blue-collar roles in used clothing CSCs, promoting sustainable development.

## 6. Conclusion

We addressed the research question posed in this study by exploring DCs of NPOs within the ever-evolving landscape of used clothing CSCs. We found that NPOs emphasized experiencing pressure stemming from impending legislations such as EPR and competition from the growing presence of for-profit entities, thus recognizing the importance of developing their capabilities to remain competitive within used clothing CSCs. This study

identifies seven capabilities pertinent to NPOs operating in used clothing CSCs. Additionally, this study goes beyond the DC approach used in previous research to detail that DCs primarily manifest at four distinct levels: *employee-level*, *managerial-level*, *organizational-level*, and *SC-level*. The framework, incorporating seven microfoundations, organized across four levels, provides a structured approach to assess an organization's current status and identify potential areas for improvement. Each level is found to have specific enablers that can support evaluation and guide ongoing development of used clothing CSCs.

Findings indicate that NPOs demonstrate capabilities in adopting profit-oriented operational and supply chain strategies to reposition themselves within the competitive landscape of used clothing CSCs, while still upholding their social responsibility. Revamping employee work integration and offering reskilling and upskilling programs aimed at fulfilling the specific skill gaps of operations such as sorting can contribute to the growth and success of used clothes CSCs. Moreover, this study emphasizes that for-profit entities must collaborate with NPOs and leverage their expertise in used clothing CSC operations, rather than attempting to replicate their efforts. This collaborative approach promises more effective and efficient management of used clothing CSCs.

### *6.1 Limitations and future research*

Our study has focused on three specific case NPOs, which may limit the generalizability of our findings. To address this limitation, future research could broaden its scope by increasing the sample size to include other geographical locations within and beyond Sweden to observe if similar patterns emerge across different contexts. Furthermore, it would be beneficial to investigate the DCs of for-profit actors operating within CSCs and conduct a comparative analysis to gain insights into the unique DCs that emerge in the two different domains. Additionally, future research should also focus on understanding enablers that facilitate the development and deployment of DCs within NPOs, particularly from the CSC perspective.

Another important direction for future research is to examine how organizations involved in used clothing CSCs develop their DCs to specifically address emerging legislative challenges such as EPR. Understanding how these organizations navigate regulatory frameworks and collaborate with stakeholders can shed light on the role of governance mechanisms in shaping CSC practices. Finally, given the growing presence of for-profit actors in used clothes CSCs, it would be valuable to explore how collaboration between NPOs and for-profit actors can facilitate the development of innovative DCs through synergies.

## References

Adro, F.D. and Fernandes, C. (2022), "Social entrepreneurship and social innovation: looking inside the box and moving out of it", *Innovation: The European Journal of Social Science Research*, Vol. 35 No. 4, pp. 704-730, doi: 10.1080/13511610.2020.1870441.

Babaei, M. and Aghdassi, M. (2022), "Measuring the dimensions of quality in service innovation: a dynamic capability and organisational competency perspective", *Total Quality Management and Business Excellence*, Vol. 33 Nos 3-4, pp. 434-466, doi: 10.1080/14783363.2020.1861933.

Bag, S. and Rahman, M.S. (2023), "The role of capabilities in shaping sustainable supply chain flexibility and enhancing circular economy-target performance: an empirical study", *Supply Chain Management*, Vol. 28 No. 1, pp. 162-178, doi: 10.1108/scm-05-2021-0246.

Bag, S., Gupta, S. and Foropon, C. (2019), "Examining the role of dynamic remanufacturing capability on supply chain resilience in circular economy", *Management Decision*, Vol. 57 No. 4, pp. 863-885, doi: 10.1108/md-07-2018-0724.

Bag, S., Gupta, S. and Luo, Z. (2020), "Examining the role of logistics 4.0 enabled dynamic capabilities on firm performance", *International Journal of Logistics Management*, Vol. 31 No. 3, pp. 607-628, doi: 10.1108/ijlm-11-2019-0311.

Bhadra, K.V., Kamalanabhan, T.J. and Singh, S.K. (2024), "Evolution of dynamic capabilities for business sustainability performance: evidence from the Indian manufacturing sector", *Business Strategy and the Environment*, Vol. 33 March, pp. 5583-5605, doi: 10.1002/bse.3767.

Bish, A. and Becker, K. (2016), "Exploring expectations of nonprofit management capabilities", *Nonprofit and Voluntary Sector Quarterly*, Vol. 45 No. 3, pp. 437-457, doi: 10.1177/0899764015583313.

Blair, E. (2015), "A reflexive exploration of two qualitative data coding techniques", *Journal of Methods and Measurement in the Social Sciences*, Vol. 6 No. 1, pp. 14-29, doi: 10.2458/v6i1.18772.

Chari, A., Niedenzu, D., Despeisse, M., Machado, C.G., Azevedo, J.D., Boavida-Dias, R. and Johansson, B. (2022), "Dynamic capabilities for circular manufacturing supply chains—Exploring the role of industry 4.0 and resilience", *Business Strategy and the Environment*, Vol. 31 No. 5, pp. 2500-2517, doi: 10.1002/bse.3040.

Chatterjee, S., Chaudhuri, R., Vrontis, D. and Thrassou, A. (2022), "Impact of organizational dynamic capability on international expansion and the moderating role of environmental dynamism", *International Journal of Organizational Analysis*, Vol. 31 No. 5, pp. 1935-1956, doi: 10.1108/IJOA-10-2021-3003.

Chowdhury, M.M.H. and Quaddus, M. (2017), "Supply chain resilience: conceptualization and scale development using dynamic capability theory", *International Journal of Production Economics*, Vol. 188 No. September 2015, pp. 185-204, doi: 10.1016/j.ijpe.2017.03.020.

Coppola, C., Vollero, A. and Siano, A. (2023), "Developing dynamic capabilities for the circular economy in the textile and clothing industry in Italy: a natural-resource-based view", *Business Strategy and the Environment*, Vol. 32 No. 7, pp. 4798-4820, doi: 10.1002/bse.3394.

Farahani, Z.R., Asgari, N. and Van Wassenhove, L.N. (2022), "Fast fashion, charities, and the circular economy: challenges for operations management", *Production and Operations Management*, Vol. 31 No. 3, pp. 1089-1114, doi: 10.1111/poms.13596.

Fisher, G. and Aguinis, H. (2017), "Using theory elaboration to make theoretical advancements", *Organizational Research Methods*, Vol. 20 No. 3, pp. 438-464, doi: 10.1177/1094428116689707.

Gupt, Y. and Sahay, S. (2015), "Review of extended producer responsibility: a case study approach", *Waste Management and Research*, Vol. 33 No. 7, pp. 595-611, doi: 10.1177/0734242x15592275.

Gustafsson, J. (2017), *Single case studies vs. multiple case studies: a comparative study"*, *Academy of Business, Engineering and Science*, Halmstad University.

Heubeck, T. (2024), "Walking on the gender tightrope: unlocking ESG potential through CEOs' dynamic capabilities and strategic board composition", *Business Strategy and the Environment*, Vol. 33 No. 3, pp. 2020-2039, doi: 10.1002/bse.3578.

Jakhar, S.K., Mangla, S.K., Luthra, S. and Kusi-Sarpong, S. (2019), "When stakeholder pressure drives the circular economy: measuring the mediating role of innovation capabilities", *Management Decision*, Vol. 57 No. 4, pp. 904-920, doi: 10.1108/md-09-2018-0990.

Khan, O., Daddi, T. and Iraldo, F. (2020), "Microfoundations of dynamic capabilities: insights from circular economy business cases", *Business Strategy and the Environment*, Vol. 29 No. 3, pp. 1479-1493, doi: 10.1002/bse.2447.

Khan, O., Daddi, T. and Iraldo, F. (2021), "Sensing, seizing, and reconfiguring: key capabilities and organizational routines for circular economy implementation", *Journal of Cleaner Production*, Vol. 287, 125565, Elsevier Ltd, doi: 10.1016/j.jclepro.2020.125565.

Kovács, G. and Spens, K.M. (2005), "Abductive reasoning in logistics research", *International Journal of Physical Distribution and Logistics Management*, Vol. 35 No. 2, pp. 132-144, doi: 10.1108/09600030510590318.

Lahane, S., Kant, R. and Shankar, R. (2020), "Circular supply chain management: a state-of-art review and future opportunities", *Journal of Cleaner Production*, Vol. 258, 120859, doi: 10.1016/j.jclepro.2020.120859.

Lu, H., Zhao, G. and Liu, S. (2022), "Integrating circular economy and industry 4.0 for sustainable supply chain management: a dynamic capability view", *Production Planning and Control*, Vol. 0 No. 0, pp. 1-17, doi: 10.1080/09537287.2022.2063198.

Lu, H., Zhao, G. and Liu, S. (2024), "Integrating circular economy and industry 4.0 for sustainable supply chain management: a dynamic capability view", *Production Planning and Control*, Vol. 35 No. 2, pp. 170-186, doi: 10.1080/09537287.2022.2063198.

Meier, O., Gruchmann, T. and Ivanov, D. (2023), "Circular supply chain management with blockchain technology: a dynamic capabilities view", *Transportation Research Part E: Logistics and Transportation Review*, Vol. 176 May, 103177, doi: 10.1016/j.tre.2023.103177.

Mishra, A., Dutta, P., Jayasankar, S., Jain, P. and Mathiyazhagan, K. (2023), "A review of reverse logistics and closed-loop supply chains in the perspective of circular economy", *Benchmarking: An International Journal*, Vol. 30 No. 3, pp. 975-1020, doi: 10.1108/bij-11-2021-0669.

Moroni, I.T., Roman Pais Seles, B.M., Lizarelli, F.L., Guzzo, D. and Mascarenhas Hornos da Costa, J. (2022), "Remanufacturing and its impact on dynamic capabilities, stakeholder engagement, eco-innovation and business performance", *Journal of Cleaner Production*, Vol. 371 July, 133274 doi: 10.1016/j.jclepro.2022.133274.

Navarro-García, A., Ledesma-Chaves, P., Gil-Cordero, E. and De-Juan-Vigaray, M.D. (2024), "Intangible resources, static and dynamic capabilities and perceived competitive advantage in exporting firms. A PLS-SEM/fsQCA approach", *Technological Forecasting and Social Change*, Vol. 198, No. November 2023, 123001, doi: 10.1016/j.techfore.2023.123001.

Nithya, R., Sivasankari, C. and Thirunavukkarasu, A. (2021), "Electronic waste generation, regulation and metal recovery: a review", *Environmental Chemistry Letters*, Vol. 19 No. 2, pp. 1347-1368, doi: 10.1007/s10311-020-01111-9.

Pal, R. and Sandberg, E. (2024), "Circular supply chain valorisation through sustainable value mapping in the post-consumer used clothing sector", *International Journal of Logistics Management*, Vol. 35 No. 5, pp. 1373-1416, doi: 10.1108/ijlm-01-2023-0023.

Panagiotopoulos, P., Protogerou, A. and Caloghirou, Y. (2023), "Dynamic capabilities and ICT utilization in public organizations: an empirical testing in local government", *Long Range Planning*, Vol. 56 No. 1, 102251, doi: 10.1016/j.lrp.2022.102251.

Patton, M.Q. (2015), *Qualitative Research and Evaluation Methods*, 4th ed., SAGE Publications, Thousand Oaks, CA.

Peng, X.E. and Liang, C. (2019), "Before nonprofit organisations become social enterprises", *Voluntas*, Vol. 30 No. 3, pp. 460-474, doi: 10.1007/s11266-018-00071-7.

Persson, O. and Hinton, J.B. (2023), "Second-hand clothing markets and a just circular economy? Exploring the role of business forms and profit", *Journal of Cleaner Production*, Vol. 390 No. June 2022, 136139, doi: 10.1016/j.jclepro.2023.136139.

Polater, A. (2021), "Dynamic capabilities in humanitarian supply chain management: a systematic literature review", *Journal of Humanitarian Logistics and Supply Chain Management*, Vol. 11 No. 1, pp. 46-80, doi: 10.1108/jhlscm-10-2020-0089.

Qiu, X., Holmen, E., Havenvid, M., De Boer, L. and Hermundsdottir, F. (2022), "Open for business: towards an interactive view on dynamic capabilities", *Industrial Marketing Management*, Vol. 107 September, pp. 148-160, doi: 10.1016/j.indmarman.2022.09.027.

Quayson, M., Bai, C., Sun, L. and Sarkis, J. (2023), "Building blockchain-driven dynamic capabilities for developing circular supply chain: rethinking the role of sensing, seizing, and reconfiguring", *Business Strategy and the Environment*, Vol. 32 No. 7, pp. 4821-4840, doi: 10.1002/bse.3395.

Rashid, A., Rasheed, R., Ngah, A.H. and Marjerison, R.K. (2024), "A nexus of multiple integrations and business performance through supply chain agility and supply flexibility: a dynamic capability view", *Journal of Science and Technology Policy Management*, Vol. ahead-of-print No. ahead-of-print, doi: 10.1108/JSTPM-08-2023-0124.

Renz, S.M., Carrington, J.M. and Badger, T.A. (2018), "Two strategies for qualitative content analysis: an intramethod approach to triangulation", *Qualitative Health Research*, Vol. 28 No. 5, pp. 824-831, doi: 10.1177/1049732317753586.

Ritola, I., Krikke, H. and Caniëls, M.C.J. (2021), "Learning-based dynamic capabilities in closed-loop supply chains: an expert study", *International Journal of Logistics Management*, Vol. 33 No. 5, pp. 69-84.

Sandberg, E. and Hultberg, E. (2021), "Dynamic capabilities for the scaling of circular business model initiatives in the fashion industry", *Journal of Cleaner Production*, Vol. 320 August, 128831, doi: 10.1016/j.jclepro.2021.128831.

Schöggl, J.P., Stumpf, L. and Baumgartner, R.J. (2024), "The role of interorganizational collaboration and digital technologies in the implementation of circular economy practices—Empirical evidence from manufacturing firms", *Business Strategy and the Environment*, Vol. 33 No. 3, pp. 2225-2249, doi: 10.1002/bse.3593.

Seuring, S. and Gold, S. (2012), "Conducting content-analysis based literature reviews in supply chain management", *Supply Chain Management*, Vol. 17 No. 5, pp. 544-555, doi: 10.1108/13598541211258609.

Shevchenko, A., Hajmohammad, S. and Pagell, M. (2024), "Operations of cost-effective charities: a qualitative study", *International Journal of Operations and Production Management*, Vol. 44 No. 5, pp. 1054-1082, doi: 10.1108/ijopm-11-2022-0755.

Stadtfeld, G.M. and Gruchmann, T. (2024), "Dynamic capabilities for supply chain resilience: a meta-review", *International Journal of Logistics Management*, Vol. 35 No. 2, pp. 623-648, doi: 10.1108/ijlm-09-2022-0373.

Teece, D.J. (2007), "Explicating dynamic capabilities:the nature and microfoundations of (sustainable) entreprise performance", *Strategic Management Journal*, Vol. 28 No. 13, pp. 1319-1350, doi: 10.1002/smj.640.

Thyer, B.A. (2010), *The Handbook of Social Work Research Methods*, Sage Publications, Thousand Oaks, London.

van Eechoud, T. and Ganzaroli, A. (2023), "Exploring the role of dynamic capabilities in digital circular business model innovation: results from a grounded systematic inductive analysis of 7 case studies", *Journal of Cleaner Production*, Vol. 401 No. April 2022, 136665, doi: 10.1016/j.jclepro.2023.136665.

Vegter, D., van Hillegersberg, J. and Olthaar, M. (2020), "Supply chains in circular business models: processes and performance objectives", *Resources, Conservation and Recycling*, Vol. 162 No. December 2019, 105046, doi: 10.1016/j.resconrec.2020.105046.

Weaven, S., Quach, S., Thaichon, P., Frazer, L., Billot, K. and Grace, D. (2021), "Surviving an economic downturn: dynamic capabilities of SMEs", *Journal of Business Research*, Vol. 128 No. June 2018, pp. 109-123, doi: 10.1016/j.jbusres.2021.02.009.

Widianto, S., Lestari, Y.D., Adna, B.E., Sukoco, B.M. and Nasih, M. (2021), "Dynamic managerial capabilities, organisational capacity for change and organisational performance: the moderating effect of attitude towards change in a public service organisation", *Journal of Organizational Effectiveness*, Vol. 8 No. 1, pp. 149-172, doi: 10.1108/joepp-02-2020-0028.

Wohlgemuth, V., Wenzel, M., Berger, E.S.C. and Eisend, M. (2019), "Dynamic capabilities and employee participation: the role of trust and informal control", *European Management Journal*, Vol. 37 No. 6, pp. 760-771, doi: 10.1016/j.emj.2019.02.005.

Yin, R.K. (2009), *Case Study Research: Design and Methods*, SAGE Publications, USA.

Yin, R.K. (2013), *Case Study Research:Design and Methods*, 3rd ed., SAGE Publications, Thousand Oaks, CA.

Zanjirani Farahani, R., Asgari, N. and Van Wassenhove, L.N. (2022), "Fast fashion, charities, and the circular economy: challenges for operations management", *Production and Operations Management*, Vol. 31 No. 3, pp. 1089-1114, doi: 10.1111/poms.13596.

Zhuravleva, A. (2024), "Reverse supply chains of non-profit organizations for textile reuse", *Journal of Humanitarian Logistics and Supply Chain Management*, Vol. 14 No. December 2023, pp. 369-383, available at: doi: 10.1108/JHLSCM-03-2023-0018.

**Corresponding author**
Kanchana Dissanayake can be contacted at: kanchana.dissanayake@hb.se

For instructions on how to order reprints of this article, please visit our website:
**www.emeraldgrouppublishing.com/licensing/reprints.htm**
Or contact us for further details: **permissions@emeraldinsight.com**